INSTITUTE OF ENGINEERING & MANAGEMENT

End Semester Examination – Odd Semester 2020

Department of Electrical and Electronics Engineering

Basic Electronics: ESC-102

Full Marks: 100 **Time: 3 hours**

Answer all questions (10x10)

1. a) Discuss the reasons for which the elemental semiconductor Silicon is preferred over the elemental semiconductor Germanium. (5)

   b) Write down a comparative study between Si and GaAs in terms of their performance as semiconductor materials. (5)

   Or

   c) Explain the energy band theory in respect of semiconductor materials with a suitable energy level diagram. (5)

   d) When a photon impinges upon a semiconductor it can take an electron from the valence band to the conduction band. The momentum is conserved in such transitions. Calculate the value of the wave vector that a 2.0 eV photon carries. The electron in the valence band can go into the conduction band with the momentum change of the photon. (5)

2. a) Discuss the effects of temperature and doping on the mobility of majority carriers in a semiconductor material. (10)

   Or

   b) What are the differences between intrinsic and extrinsic semiconductors? (5)

   c) How the valency of an external impurity affects the nature of a semiconductor? (5)

3. a) Explain the process of diffusion for the carriers of a semiconductor materials. How it is related to the drift velocity? (10)

   Or

   b) Explain any two types of common electron-hole recombination phenomenon inside a semiconductor material. (10)

4. a) Explain the principle of operation for a p-n junction diode in terms of energy band diagrams. (10)

   Or

   b) Discuss the effects of doping concentration on the barrier potential for a p-n junction diode. (10)

5. a) Explain the phenomenon of Zener breakdown for a semiconductor diode with suitable V-I characteristics and write down a comparative study of this with avalanche breakdown. (10)

Or

b) What do you mean by Avalanche multiplication? (5)

c) What is PIV and threshold voltage of a diode? (5)

6. a) Write down the construction and working principle of a photovoltaic diode. (5)

b) Briefly discuss about the working principle of a light emitting diode. (5)

Or

c) The avalanche diode regulates at 60 V over a range of diode currents from 5 to 40 mA. For a supply voltage V = 200 Volt, calculate the value of R to allow voltage regulation from a load current I_L = 0up to I_{max}, the maximum possible value of I_L. What is the value of I_{max}? If R is set as calculated above and the load current is set at I_L = 25 mA, what are the limits between which V may vary without loss of regulation in the circuit? (10)

7. a) For a p-n-p transistor, explain its principle of operation with the help of a suitable schematic diagram. (7)

b) Why the thickness of the base region is minimum for a BJT. (3)

Or

c) Show the circuit configuration for a common-collector mode of operation for a BJT. (3)

d) Derive the relationship between α and β for a BJT, where the symbols carry their usual meaning. (7)

8. a) Draw and explain the input characteristics for a BJT both for common-emitter and common-base modes. (10)

Or

b) Write down a comparative study between BJT and FET. (5)

c) Compare V-I characteristics of BJT and FET. (5)

9. a) When used in the common-base configuration mode, a transistor with α = 0.90 gives a reverse saturation current I_{CO} = 15 μA. Calculate the collector current when the transistor is connected in the common-emitter mode with a base current of 0.5mA. (10)

Or

b) For a BJT having a base current of 50 μA and a collector current of 5 mA, what is the value of emitter current? What is β for this transistor using the computed value of I_E? Also calculate the value of the common-base current gain α. (10)

10. a) Draw and explain the basic configuration and working principle of an n-channel depletion type MOSFET. (10)

Or

b) With the help of its output characteristics, explain how can a BJT be operated in different modes. (10)

INSTITUTE OF ENGINEERING & MANAGEMENT

Mid Semester Examination – Odd Semester 2020

Department of Basic Science & Humanities

Basic Electronics: ESC-102

Full Marks: 100 **Time: 3 hours**

Answer all questions (10x10)

1. a) What do you mean by valence band and conduction band? (5)
   b) Write the basic criterion for the classification of metals, semiconductors and insulators. (5)

   Or

   c) How junction capacitance is formed in P-N junction diode? (5)
   d) What do you mean by threshold voltage? (5)

2. a) A donor impurity is added to an intrinsic semiconductor and resistivity at room temperature is 9.6 ohm-cm. Calculate the donor atom density if mobility of electrons is 1300 cm^2/ Vsec. (10)

   Or

   b) Draw the Fermi level diagram for P type and N type semiconductor. (5)
   c) Elaborate the difference between drift current and diffusion current. (5)

3. a) Calculate the fraction of drift current due to holes to the drift current due to electrons in intrinsic semiconductor if mobility of electrons is 3800 cm^2 / V sec and mobility of holes is 1800 cm^2/ Vsec. (10)

   Or

   b) Derive the expression for current density and conductivity in semiconductor. (10)

4. a) In an n-type semiconductor at T=300K, the electron concentration varies linearly from 10^{17} to $3*10^{16}/cm^3$ over a distance of 0.72 cm. Calculate the diffusion current density if the electron diffusion co efficient is 268 cm^2/s. (10)

   Or

   b) Write down the difference between Zener breakdown and avalanche breakdown. (5)
   c) Explain how the avalanche multiplication occurs? (5)

5. a) Draw and explain the characteristic of PN junction diode and label the graph properly for forward bias and reverse bias. (10)

   Or

b) When a p-n junction is said to be forward-biased and reverse-biased? Explain with figure. (10)

6. a) Write down the mass action law. (5)
b) What do you mean by transistor amplification factor? (5)

Or

c) An abrupt Si p-n diode at 300K has a doping of $N_a = 10^{18}/cm^3$. Calculate the value of N_d. Given intrinsic carrier concentration $n_i = 10^{10}/cm^3$ (10)

7. a) What is space charge region? (5)
b) What is the effect of doping on barrier potential? (5)

Or

c) Write down the difference between Intrinsic and extrinsic semiconductor? (5)
d) Why at 0K intrinsic semiconductor behaves like an insulator? (5)

8. a) What is the origin of the name transistor? (5)
b) Establish the relation between α and β. (5)

Or

c) Define the term recombination and generation. (5)
d) What do you mean by transistor load line? Explain its significance. (5)

9. a) Draw the band diagram of p-type and n-type semiconductor? (5)
b) Write the differences between metal and insulators. (5)

Or

c) An NPN BJT having reverse saturation current $I_S = 10-15$ A is biased in the forward active region with $V_{BE} = 700$ mV and the current gain (β) may vary from 50 to 150 due to manufacturing variations. Calculate the maximum emitter current? (10)

10. a) Draw common base input characteristics of a transistor. (5)
b) Write down the difference between BJT and FET. (5)

Or

c) Describe with diagram the process of p-n junction formation in detail. (10)

INSTITUTE OF ENGINEERING & MANAGEMENT

End Semester Examination – Odd Semester 2020

Department of Basic Science & Humanities

Basic Electronics: ESC-102

Full Marks: 100 **Time: 3 hours**

Answer all questions (10x10)

1. a) Write the differences between Crystalline and Amorphous materials? (5)
   b) Draw the band diagram of p-type and n-type semiconductor. (5)

   Or

   c) Write the differences between metal and insulators. (5)
   d) What is space charge region? (5)

2. a) How junction capacitance is formed in P-N junction diode? (5)
   b) What do you mean by built in potential? (5)

   Or

   c) Describe the band diagram of p-n junction diode with proper diagram? (10)

3. a) Evaluate the expression of Drift and Diffusion current. (10)

   Or

   b) An acceptor impurity is added to an intrinsic semiconductor and resistivity at room temperature is 9.6 ohm cm. Calculate the donor atom density if mobility of holes is 1300 cm^2/Vsec. (10)

4. a) Explain with diagram the I-V characteristic of p-n junction diode. (5)
   b) Give one application of Zener diode. (5)

   Or

   c) What is biasing? How many types of biasing are there? (5)
   d) Describe how p-n junction diode behaves as switch? (5)

5. a) How many types of breakdown can occur in p-n junction? Write the differences between them with diagram. (10)

   Or

   b) In an n-type semiconductor at T=300K, the electron concentration varies linearly from 10^{17} to $3*10^{16}/cm^3$ over a distance of 0.72 cm. Calculate the diffusion current density if the electron diffusion co efficient is 268 cm^2/s. (10)

6. a) An abrupt Si p-n diode at 300K has a doping of $N_a = 10^{18}/cm^3$. Calculate the value of N_d. Given intrinsic carrier concentration $n_i = 10^{10}/cm^3$ (10)

Or

b) An electric field of 137V/cm is applied across an n type semiconductor sample and the minority carriers are being injected to the sample at a point. The field moves these carriers a distance of 0.4 cm in 10 μs. Find the drift velocity and the diffusivity of the minority carriers. (10)

7. a) Write the differences between BJT and FET. (5)
b) Explain with diagram what is emitter follower circuit? (5)

Or

c) Draw the input and output characteristics of CE amplifier and mention the active, saturation and cut off region in the graph. (5)
d) What is operating point? In which region BJT is used as switch? (5)

8. a) Write down the difference between Intrinsic and extrinsic semiconductor? (5)
b) Why at 0K intrinsic semiconductor behaves like an insulator? (5)

Or

c) Write the differences between depletion type MOSFET and enhancement type MOSFET. (5)
d) What is threshold voltage? (5)

9. a) Define depletion width and with a rough sketch show their formation in different operating modes of PN junction diodes. (10)

Or

b) Draw the circuit diagram of CE amplifier with BJT and describe the reason for bypass capacitors. (10)

10. a) Define Fermi energy level and band gap of semiconductor. (5)
b) Define the cut-in voltage and knee voltage with diagram. (5)

Or

c) Explain the mechanism of Avalanche breakdown in pn junction. (10)